HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and debugged embedded C++ firmware for **real-time operating system** environments, ensuring stability for core platform software.
- Designed and implemented **scalable build system software** and tooling to automate deployment pipelines and streamline developer workflows.
- Collaborated with **HW and Systems Engineering** teams to integrate peripheral software and validate power and performance KPIs against product specifications.
- Performed **image and system-level sanity testing** on new firmware releases, identifying and documenting critical defects for resolution.
- Wrote **Python scripts** to automate data extraction from system logs for root cause analysis of integration issues, reducing manual effort by approximately 40%.
- Managed **software source control** using Git for a team of 5 engineers, implementing branching strategies and conducting code reviews for feature integration.
- Supported production environments by analysing system logs and debugging issues in **Linux-based** infrastructure, contributing to incident response.

SKILLS

Languages – C, C++, Assembly, Python, SQL

Embedded Systems – Real-time operating systems, Firmware Development, Driver Integration, Peripheral Software

Development Tools – Build Systems, Software Source Control (Git), Debugging Tools, Jenkins

Operating Systems – Linux, RTOS

Methodologies – Agile, Scrum, Integration Testing, System-Level Testing

PROJECTS

Firmware Integration for IoT Device Fleet

- Integrated and tested embedded C/C++ firmware modules for a fleet of **IoT devices**, focusing on reliable data transmission.
- Developed **Python automation scripts** to simulate device network traffic and validate integration points, catching over 50% of integration defects pre-QA.
- Managed firmware version control and coordinated build deployments for 3 distinct device firmware releases over a 6-month period.

Automated Build and Deployment Pipeline

- Engineered a CI/CD pipeline using **Jenkins** and **Git** to automate the build, test, and deployment of embedded software components.
- Implemented **scripting in Perl** to parse build logs and generate automated test reports, improving defect detection timelines by 30%.
- Collaborated with system architects to define build configurations and ensure adherence to **power and performance KPIs** for the core platform.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)